

Department of Computer Science and Engineering

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BUSINESS ANALYTICS

Case Study Report

**Analysis of Student Willingness-to-Pay for Premium Ed-Tech Tools:
A Predictive Modelling Approach**

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Table of Contents

Contents

1	Problem Statement and Objectives	1
1.1	Problem Statement	1
1.2	Objectives	1
2	Data Collection and Dataset Description	1
2.1	Data Source	1
2.2	Data Collection Procedure	2
2.3	Dataset Distribution	2
2.4	Dataset Attributes	2
2.5	Dataset Limitations	3
3	Data Preparation and Exploratory Analysis	3
3.1	Data Cleaning	3
3.2	Missing Values	4
3.3	Rating Distribution	4
3.4	Review Length Analysis	4
3.5	Premium-Related Feature Engineering	5
3.6	Premium Engagement Distribution	6
3.7	Application-Level Analysis	6
3.8	Rating and Premium Engagement	7
4	Analytics Method and Implementation	7
4.1	Analytical Objective	7
4.2	Predictor Variables	7
4.3	Train-Test Split	8
4.4	Preprocessing	8
4.5	Predictive Models	8
4.6	Evaluation Metrics	8
4.6.1	Mean Absolute Error	9
4.6.2	Root Mean Squared Error	9
4.6.3	Coefficient of Determination	9
5	Comparison with State-of-the-Art Methods	9
5.1	Comparison with the Present Study	10
6	Results, Business Insights and Recommendations	11
6.1	Model Results	11
6.2	Model Error Comparison	11
6.3	R^2 Comparison	12
6.4	Model Interpretation	12
6.5	Business Insights	12
6.6	Business Recommendations	13
7	Conclusion and References	14
7.1	Conclusion	14
7.2	References	14
A	Feature Summary	15
B	Reproducibility Structure	15

Abstract

Ed-Tech platforms increasingly use freemium business models in which basic functionality is provided at no cost while advanced features are offered through premium subscriptions. Although such models attract large numbers of users, identifying users who demonstrate meaningful interest in premium features remains a business analytics challenge. This study investigates review-based signals associated with premium engagement in Ed-Tech applications using publicly available app reviews.

A dataset of 10,928 unique reviews from Duolingo, Photomath, and Brainly was analysed. The dataset was collected through a structured public app-review dataset API and contains review ratings, review text, application information, helpfulness indicators, and review metadata. Duplicate records were removed and review-level textual indicators were engineered to identify references to premium subscriptions, purchasing, pricing, free access, cancellation, and feature limitations.

A review-derived premium engagement score was created as a proxy measure rather than as a direct observation of willingness-to-pay or actual subscription conversion. Linear Regression, Decision Tree Regression, and Random Forest Regression were developed using review rating, review length, word count, helpfulness, and application as predictors. Linear Regression obtained an MAE of 0.1317, RMSE of 0.3126, and R^2 of 0.1645. Decision Tree Regression obtained an MAE of 0.1174, RMSE of 0.3204, and R^2 of 0.1222, while Random Forest obtained an MAE of 0.1190, RMSE of 0.3226, and R^2 of 0.1106.

The findings indicate that public review metadata can provide useful exploratory insights into premium-related user discussions, but the available variables explain only a limited portion of the variation in the proxy engagement score. The study therefore highlights the value of combining review analytics with transaction, demographic, subscription, and behavioural data for more direct future conversion modelling.

1 Problem Statement and Objectives

1.1 Problem Statement

Ed-tech platforms attract millions of users through freemium models, but only a small percentage convert to premium subscriptions, resulting in inefficient marketing efforts and lost revenue opportunities. Businesses need a data-driven approach to identify users with genuine purchase intent rather than relying solely on basic engagement metrics.

This study uses publicly available app-review text and platform-level review metadata to identify observable signals associated with premium engagement, while recognising that verified behavioural, attitudinal, demographic, and transaction-level information is unavailable.

In the present study, publicly available app reviews are used as an observable source of user feedback. Since the dataset does not contain verified subscription transactions, demographic information, or direct willingness-to-pay responses, the analysis focuses on review-derived premium engagement signals as a proxy.

1.2 Objectives

The main objectives of the case study are:

1. To collect and analyse a large publicly available dataset of Ed-Tech application reviews.
2. To identify review-based signals associated with premium features, pricing, subscriptions, free access, cancellation, and feature limitations.
3. To develop and evaluate predictive models for the review-derived premium engagement score and derive practical business insights for premium-feature and subscription strategy.

2 Data Collection and Dataset Description

2.1 Data Source

The dataset consists of publicly available application reviews from three Ed-Tech applications:

- Duolingo
- Photomath
- Brainly

The reviews were obtained through the Crawlora structured App Reviews Dataset API. The API provides stored structured records of publicly available app reviews, including review text, rating, application identifier, country, review date, helpfulness information, and other metadata.

The source was selected because it provides a structured mechanism for collecting a large number of public app-review records required for the case study.

The data source documentation was consulted to understand the available fields and dataset retrieval procedure.

2.2 Data Collection Procedure

The data collection procedure consisted of the following steps:

1. The public app-review dataset endpoint was accessed through the Crawlora API.
2. Review records were collected for Duolingo, Photomath, and Brainly.
3. Multiple API queries were used to obtain sufficient records for the required dataset size.
4. The collected records were combined into a single dataset.
5. Duplicate reviews were identified using the review identifier.
6. Duplicate records were removed before analysis.
7. The final cleaned dataset contained 10,928 unique review records.

A total of 11,200 raw records were collected. After removing 272 duplicate records, 10,928 unique reviews remained for analysis.

2.3 Dataset Distribution

The final dataset contains:

Table 1: Distribution of Reviews by Application

Application	Reviews	Percentage
Duolingo	3,928	35.95%
Photomath	3,500	32.02%
Brainly	3,500	32.02%
Total	10,928	100%

2.4 Dataset Attributes

The main attributes used in the analysis are:

Table 2: Important Dataset Attributes

Attribute	Description
app	Name of the application associated with the review.
score	User rating assigned to the application, from 1 to 5.
text	Textual content of the user review.
review_id	Identifier associated with the review.
country	Country associated with the review record.
version	Application version when available.
thumbs_up	Helpfulness indicator associated with the review.
reviewed_at	Review date.

2.5 Dataset Limitations

The dataset represents public app-review activity and does not provide verified student identities, demographic characteristics, subscription transactions, actual payment information, or confirmed willingness-to-pay values.

Therefore, the analysis does not claim that every reviewer is a student or that a premium-related review necessarily resulted in a purchase. Instead, review language is used to construct an observable proxy for premium engagement.

3 Data Preparation and Exploratory Analysis

3.1 Data Cleaning

The following preprocessing steps were performed:

- Duplicate review records were removed.
- Missing review text was retained as an empty textual value where necessary.
- Numerical variables were inspected for missing values.
- Review text was converted to a consistent format for keyword-based feature extraction.
- Text length and word count were calculated.
- Premium-related keyword indicators were generated.

After cleaning, the final dataset contained 10,928 unique reviews.

3.2 Missing Values

The missing-value analysis identified the following major patterns:

Table 3: Missing Values in the Dataset

Attribute	Missing Values
title	10,928
text	1
version	1,949
thumbs_up	270
Other major fields	0

The title field was not used as a predictive variable because it was unavailable for the collected records.

3.3 Rating Distribution

The distribution of review ratings was:

Table 4: Review Rating Distribution

Rating	Count	Percentage
1	1,943	17.79%
2	282	2.58%
3	472	4.32%
4	1,151	10.54%
5	7,080	64.78%

The average rating across the complete dataset was approximately 4.02.

3.4 Review Length Analysis

Review text length was calculated using the number of characters and words.

The character-length statistics were:

Table 5: Review Length Statistics

Statistic	Value
Mean	60.03 characters
Standard deviation	86.11
Minimum	0
Median	28
75th percentile	67
Maximum	500

3.5 Premium-Related Feature Engineering

The study initially considered direct keyword-based purchase-intent classification. However, direct purchase-intent labels were extremely sparse in the collected reviews.

Therefore, instead of using a highly imbalanced binary purchase label, a broader review-derived premium engagement score was created.

Six categories of premium-related signals were extracted:

- Premium and subscription references
- Purchase and payment references
- Price and affordability references
- Free access and advertisements
- Cancellation and refund references
- Limitations and locked-feature references

The corresponding binary indicators were created using keyword matching.

Table 6: Premium-Related Text Signals

Feature	Reviews Containing Signal
has_premium	167
has_purchase	224
has_price	39
has_free	377
has_cancel	29
has_limitation	34

The premium engagement score was calculated as:

$$\text{PremiumEngagementScore} = \text{Premium} + \text{Purchase} + \text{Price} + \text{Free} + \text{Cancel} + \text{Limitation}$$

The resulting score was then divided into three engagement levels:

- Low: score = 0
- Medium: score = 1
- High: score ≥ 2

3.6 Premium Engagement Distribution

The final engagement distribution was:

Table 7: Premium Engagement Distribution

Engagement Level	Count	Percentage
Low	10,274	94.02%
Medium	491	4.49%
High	163	1.49%

The average premium engagement score was approximately 0.080.

3.7 Application-Level Analysis

The application-level analysis showed:

Table 8: Application-Level Premium Engagement

Application	Reviews	Avg. Score	High Engagement
Brainly	3,500	0.027	12
Duolingo	3,928	0.113	88
Photomath	3,500	0.095	63

The corresponding high-engagement proportions were approximately:

- Brainly: 0.34%
- Duolingo: 2.24%
- Photomath: 1.80%

3.8 Rating and Premium Engagement

Premium-related engagement was more frequently observed among lower-rated reviews.

For example, 1-star reviews contained:

- 84.20% Low engagement
- 10.45% Medium engagement
- 5.35% High engagement

In comparison, 5-star reviews contained:

- 97.88% Low engagement
- 1.85% Medium engagement
- 0.27% High engagement

This pattern indicates that premium-related discussion appears more frequently in negative reviews.

4 Analytics Method and Implementation

4.1 Analytical Objective

The predictive modelling component estimates the review-derived premium engagement score using observable review metadata and application information.

The target variable is:

$$Target = PremiumEngagementScore$$

The target is a proxy constructed from review language and should not be interpreted as an observed subscription transaction or direct willingness-to-pay measure.

4.2 Predictor Variables

The following predictor variables were used:

- Review rating
- Review length
- Word count

- Thumbs-up count
- Application

The keyword indicators used to construct the target were excluded from the predictive feature set to avoid direct target leakage.

4.3 Train-Test Split

The dataset was divided into training and testing subsets using an 80:20 split.

Table 9: Train-Test Split

Dataset	Records
Training set	8,742
Testing set	2,186
Total	10,928

A random state of 42 was used to make the split reproducible.

4.4 Preprocessing

Numerical features were processed using median imputation.

The application variable was processed using:

- Most-frequent-value imputation
- One-hot encoding

The preprocessing steps were implemented using a Scikit-learn pipeline.

4.5 Predictive Models

Three regression models were evaluated:

1. Linear Regression
2. Decision Tree Regression
3. Random Forest Regression

4.6 Evaluation Metrics

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2).

4.6.1 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE measures the average absolute difference between the actual and predicted values.

4.6.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE gives greater weight to larger prediction errors.

4.6.3 Coefficient of Determination

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

R^2 indicates the proportion of variation in the target that is explained by the model.

5 Comparison with State-of-the-Art Methods

Recent research demonstrates different approaches for analysing willingness-to-pay, subscription-payment intention, and student perspectives toward digital education.

Table 10: Comparison with Recent Published Studies

Study / Year	Dataset	Method	Evaluation Metric	Key Result	Comparison with This Work
Tyrväinen & Karjaluo (2024)	Meta-analytical data from 55 studies and 15 qualitative interviews on freemium services	Mixed-method research and meta-analysis	Meta-analytic effect sizes and qualitative analysis	Perceived functional, hedonic, social, and price value were identified as predictors of willingness-to-pay. Trust also mediated relevant relationships.	Directly relevant to freemium and willingness-to-pay. The present study instead uses large-scale public app reviews and a review-derived proxy.

Study / Year	Dataset	Method	Evaluation Metric	Key Result	Comparison with This Work
Veguru, Naren & Singam (2024)	Questionnaire responses from college students regarding online education	Machine-learning classification	Classification accuracy	Random Forest achieved a reported accuracy of 93% among the tested classification approaches.	Uses student-focused educational data and ML classification, whereas the present study uses public reviews and regression modelling.
Liu, Park & Wang (2025)	274 responses from ChatGPT users	Regression and moderated mediation analysis	Reliability, validity, regression, and moderated mediation analysis	Perceived usefulness positively influenced satisfaction and subscription-payment intention; satisfaction partially mediated the relationship.	Directly examines subscription-payment intention using survey data, whereas the present study uses 10,928 public reviews and a text-derived proxy.

5.1 Comparison with the Present Study

The present study differs from these studies mainly in its data source and target definition.

Instead of questionnaire responses, verified subscription transactions, or detailed learning-management-system clickstream records, this study uses 10,928 publicly available app reviews from Duolingo, Photomath, and Brainly.

The study constructs a review-derived `premium_engagement_score` from observable references to premium subscriptions, payment, price, free access, cancellation, and feature limitations.

This score provides a scalable mechanism for analysing public user feedback, but it is not equivalent to observed subscription purchase behaviour or directly measured willingness-to-pay.

Unlike studies using transaction or learner-level behavioural data, the present models use basic review metadata and text-derived characteristics while excluding the keyword indicators used to construct the target.

The relatively limited R^2 values indicate that the available predictors explain only a small portion of the variation in the proxy target.

The comparison therefore highlights the importance of richer behavioural, demographic, survey, and transaction-level information when the objective is direct prediction of subscription payment or willingness-

to-pay.

6 Results, Business Insights and Recommendations

6.1 Model Results

The predictive modelling results are shown below.

Table 11: Predictive Model Performance

Model	MAE	RMSE	R^2
Linear Regression	0.1317	0.3126	0.1645
Decision Tree	0.1174	0.3204	0.1222
Random Forest	0.1190	0.3226	0.1106

The results show that the models have relatively limited explanatory power for the review-derived premium engagement score.

6.2 Model Error Comparison

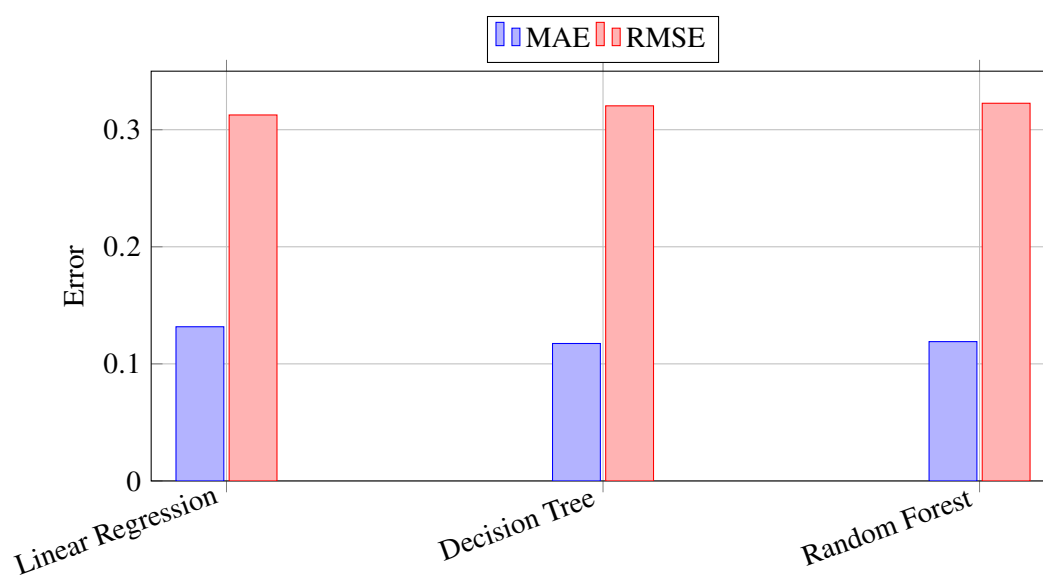


Figure 1: Comparison of MAE and RMSE across predictive models

6.3 R^2 Comparison

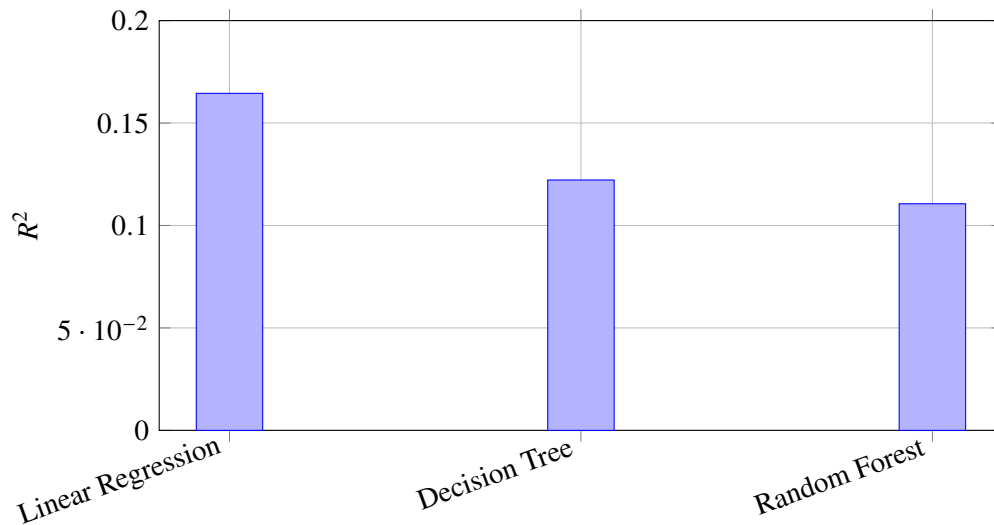


Figure 2: R^2 comparison across predictive models

6.4 Model Interpretation

The three regression models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 .

Linear Regression produced an MAE of 0.1317, RMSE of 0.3126 and R^2 of 0.1645. The Decision Tree produced an MAE of 0.1174, RMSE of 0.3204 and R^2 of 0.1222, while the Random Forest produced an MAE of 0.1190, RMSE of 0.3226 and R^2 of 0.1106.

The results indicate that the available review metadata and basic text-length characteristics explain only a limited portion of the variation in the review-derived premium engagement score. Therefore, the models should be interpreted as exploratory predictive models rather than accurate predictors of actual subscription conversion.

The relatively limited R^2 values are also consistent with the limitations of the dataset. The target is a proxy constructed from review language rather than observed transaction data, while demographic, behavioural, and verified subscription information is unavailable.

6.5 Business Insights

The exploratory analysis provides several patterns relevant to Ed-Tech subscription and premium-feature strategy.

- Premium-related engagement is concentrated in a small proportion of reviews. Most reviews contain no detected premium-related signal, while 163 reviews have two or more signals.
- Duolingo has the highest average premium engagement score among the three applications, followed by Photomath and Brainly.

- Premium-related engagement is more frequently observed among lower-rated reviews. Reviews with a rating of 1 contain a higher proportion of medium and high premium engagement than 4- and 5-star reviews.
- Review language frequently discusses subscriptions, payment requirements, advertisements, free-tier restrictions, and feature limitations. These themes provide useful signals for understanding how users discuss premium experiences.
- The predictive models achieved relatively limited R^2 values. This indicates that basic review metadata alone does not explain most variation in the review-derived premium engagement score.

From a business analytics perspective, the findings suggest that Ed-Tech platforms can use review text and rating patterns as an exploratory feedback-monitoring mechanism. However, transaction-level subscription data and verified user characteristics would be required for reliable individual-level conversion prediction.

6.6 Business Recommendations

Based on the exploratory analysis and predictive modelling results, the following recommendations can be considered by Ed-Tech platforms:

1. Monitor premium-related user feedback

Platforms can systematically analyse reviews for references to subscriptions, pricing, advertisements, feature restrictions, and premium access. This can help identify recurring concerns related to monetization.

2. Investigate negative premium experiences

Lower-rated reviews showed a higher proportion of premium-related engagement. Platforms can examine these reviews to identify whether pricing, advertisements, restrictions, or locked features are contributing to dissatisfaction.

3. Use application-specific feedback

The three applications showed different levels of premium engagement. Subscription and feature strategies should therefore consider platform-specific user feedback rather than applying the same approach across applications.

4. Combine review data with behavioural data

Review data alone provided limited predictive power in this study. Combining review signals with variables such as usage frequency, subscription history, feature usage, demographics, and transaction information could provide a stronger basis for conversion prediction.

5. Use predictive models as decision-support tools

The models developed in this study should be treated as exploratory decision-support tools rather than direct predictors of actual willingness-to-pay. Business decisions should incorporate additional verified customer and transaction-level information.

7 Conclusion and References

7.1 Conclusion

This study analysed publicly available app reviews from Duolingo, Photomath, and Brainly to investigate user discussions related to premium features, pricing, and subscription experiences. A total of 10,928 unique reviews were analysed after removing duplicate records.

The exploratory analysis showed that premium-related engagement was concentrated in a relatively small proportion of reviews. Duolingo recorded the highest average premium engagement score among the three applications, followed by Photomath and Brainly. Lower-rated reviews also contained premium-related signals more frequently, suggesting that pricing, advertisements, restrictions, and premium feature access can be important themes in negative user feedback.

Three regression models were developed using review rating, review length, word count, helpfulness, and application as predictors of the review-derived premium engagement score. Linear Regression achieved an R^2 of 0.1645, while Decision Tree and Random Forest achieved R^2 values of 0.1222 and 0.1106 respectively.

These results indicate that basic review metadata and text characteristics explain only a limited portion of variation in the proxy target.

The study demonstrates that publicly available review data can provide useful exploratory insights for Ed-Tech subscription analytics. However, the premium engagement score is a proxy and should not be interpreted as actual willingness-to-pay or confirmed subscription conversion.

Future work could incorporate verified transaction data, user demographics, subscription history, usage frequency, and detailed behavioural data to build more direct and reliable conversion prediction models.

Overall, the analysis provides a data-driven framework for monitoring premium-related user feedback and identifying themes that can support future pricing, feature-gating, and customer-engagement decisions.

7.2 References

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A Feature Summary

The main engineered features used in the analysis are:

Table 12: Feature Engineering Summary

Feature	Description
review_length	Number of characters in the review text.
word_count	Number of words in the review text.
exclamation_count	Number of exclamation marks.
question_count	Number of question marks.
has_premium	Indicates premium/subscription-related terms.
has_purchase	Indicates purchase/payment-related terms.
has_price	Indicates pricing or affordability terms.
has_free	Indicates free-access, advertisement, or freemium terms.
has_cancel	Indicates cancellation, refund, or unsubscribe terms.
has_limitation	Indicates limitation, restriction, or locked-feature terms.
premium_engagement_score	Sum of the six premium-related binary indicators.
premium_engagement_level	Low, Medium, or High based on the engagement score.

B Reproducibility Structure

The project is organised as follows:

```
EdTech_Case_Study/
|
|-- README.md
|
|-- data/
|   |-- raw/
|   |
|   '-- processed/
|       |-- edtech_reviews_10928_unique.csv
|       |-- edtech_reviews_features.csv
|       '-- edtech_reviews_analysis.csv
```

```
|  
|-- notebooks/  
|   '-- analysis.ipynb  
|  
|-- report/  
    '-- Case_Study_Report.pdf
```